

Course Syllabus

Frontend Development with React (Project-Based)

Course Objective

This course is designed to provide a strong foundation in modern web development using React. Learners will understand core web concepts, React internals, state management, API integration, and professional tooling by building a project.

Module 1 : Web Fundamentals

- How websites work
- Client–Server Architecture
- HTTP vs HTTPS
- Request–Response lifecycle
- HTTP methods (GET, POST, PUT, DELETE)
- HTTP status codes (2xx, 3xx, 4xx, 5xx)
- APIs and API types
- REST vs RESTful APIs
- Role of databases in web applications

Module 2 : Introduction to React

- What is React and why it is used
- Problems React solves in modern web development
- Single Page Applications (SPA)
- Component-based architecture
- JSX syntax and rules
- JSX transpilation (Babel overview)

Module 3. How React Works Internally ?

- Virtual DOM
- Diffing algorithm
- Reconciliation process
- Rendering and re-rendering
- State-driven UI updates

Module 4 : React Project Setup , Hooks ,Routing

- Functional components
- Props
- State management with useState
- Why useState is preferred over variables.
- Event handling
- Conditional rendering
- Controlled vs uncontrolled components
- React Router fundamentals
- Static and dynamic routes
- Protected Routes

Module 5 : API Integration & Asynchronous JavaScript

- Promises, callbacks, and async/await
- Fetch API
- Error handling strategies
- Mock APIs and CRUD simulation
- API-driven UI development

Module 6: Project Development (Basics)

- UI development using Tailwind CSS
- Reusable components (Navbar, Cards, Footer)
- Home page,Detail page with dynamic routing
- Create, edit, and delete functionalities.
- State-lifting concept.
- Search functionality

Module 7: Project Development (Advanced)

- React Hook Form
- Introduction to Axios
- Axios instance configuration
- Why global state is required
- Introduction to Redux Toolkit
- Store configuration
- Slices and reducers
- Dispatching actions and selecting state
- Managing authentication and authorization
- Environment variables

Module 8: Version Control & Deployment

- Git fundamentals
- Deploying React applications on Vercel

Learning Outcomes

By the end of this course, learners will be able to:

- Understand modern web architecture and APIs
- Build scalable React applications
- Integrate backend services efficiently
- Efficiently Utilize professional tools like React Hook Form, Axios...
- Deploy production-ready applications
- Follow industry best practices and workflows